

## Finding Orthologs and Paralogs

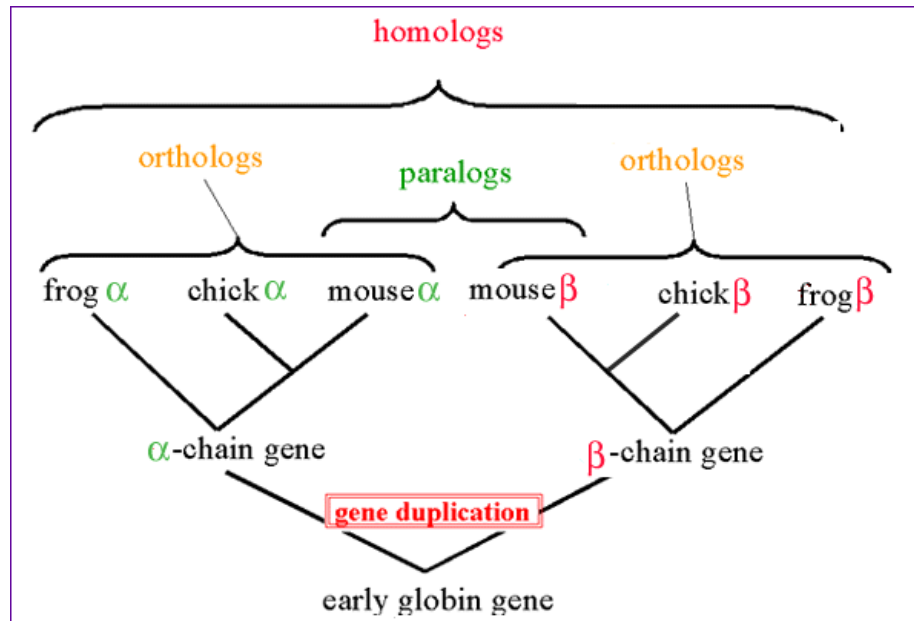


Figure 1: Orthologs

### Gene families and Orthology

Problem: How to find “same” genes across multiple species.

Genes can duplicate (Paralogs) and can be identical due to descent (Ortholog)

### Methods

- BLAST: 1 way BLAST (Gene A in Species X, what is best hit in Species Y)
- BLAST: reciprocal BLAST

### Trees can help resolve relationships

Best hits can sometimes be wrong (B) though it can be resolved with phylogenetics.

### Reciprocal Searches

- Bi-directional or Reciprocal BLAST

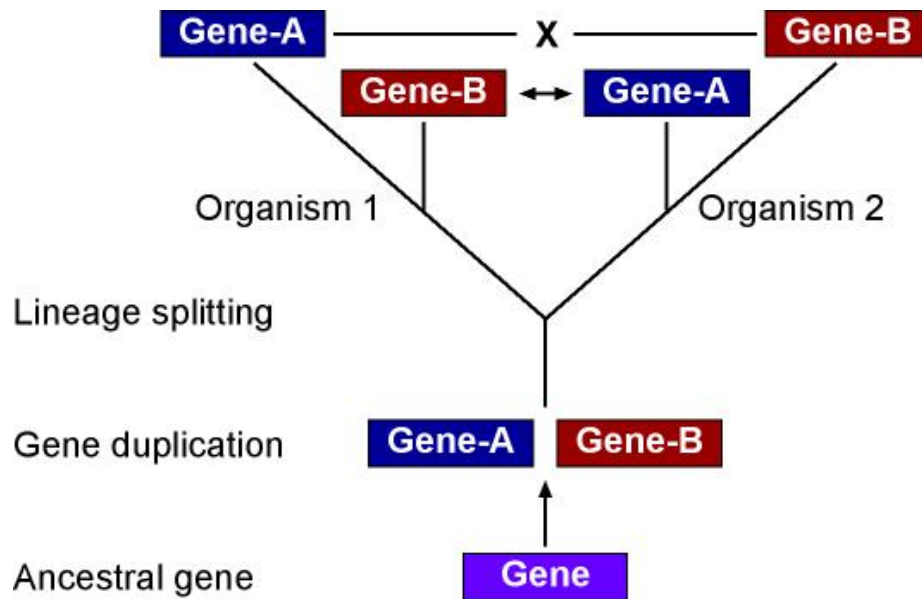


Figure 2: orthologs

## Implement Bidirectional

Method to find best top hit in one direction and the reverse.

Let's walk through the [code](#)

*Will write this in Python in Class*

## Clustering

- Lumping genes together based on similarity linkage
- Single-linkage means if there is a link between A-B then they are in a cluster

## Code up single-linkage

Let's look at some [code](#).

*Will write this in Python in Class*

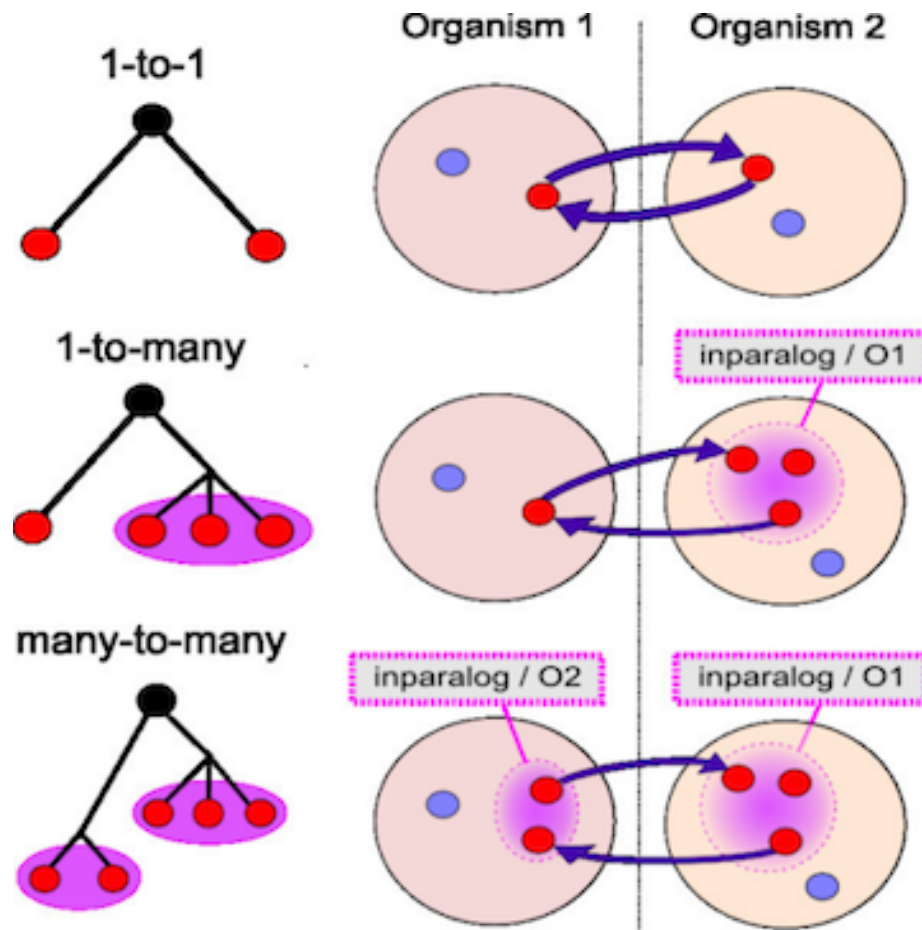


Figure 3: diagramorth

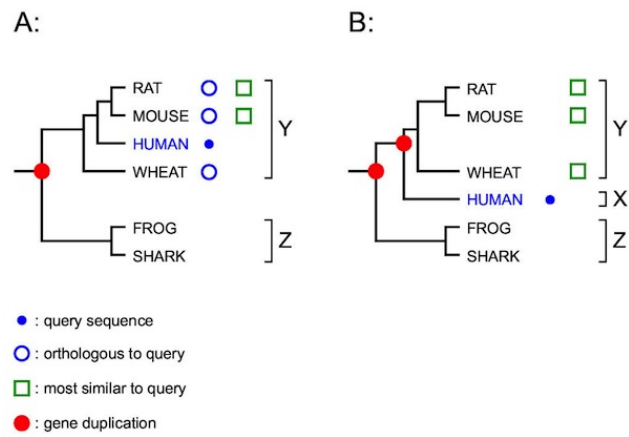


Figure 4: RIO

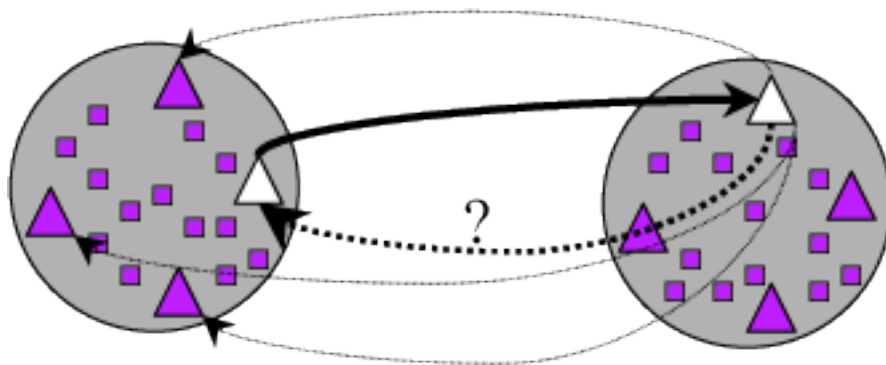


Figure 5: BRH

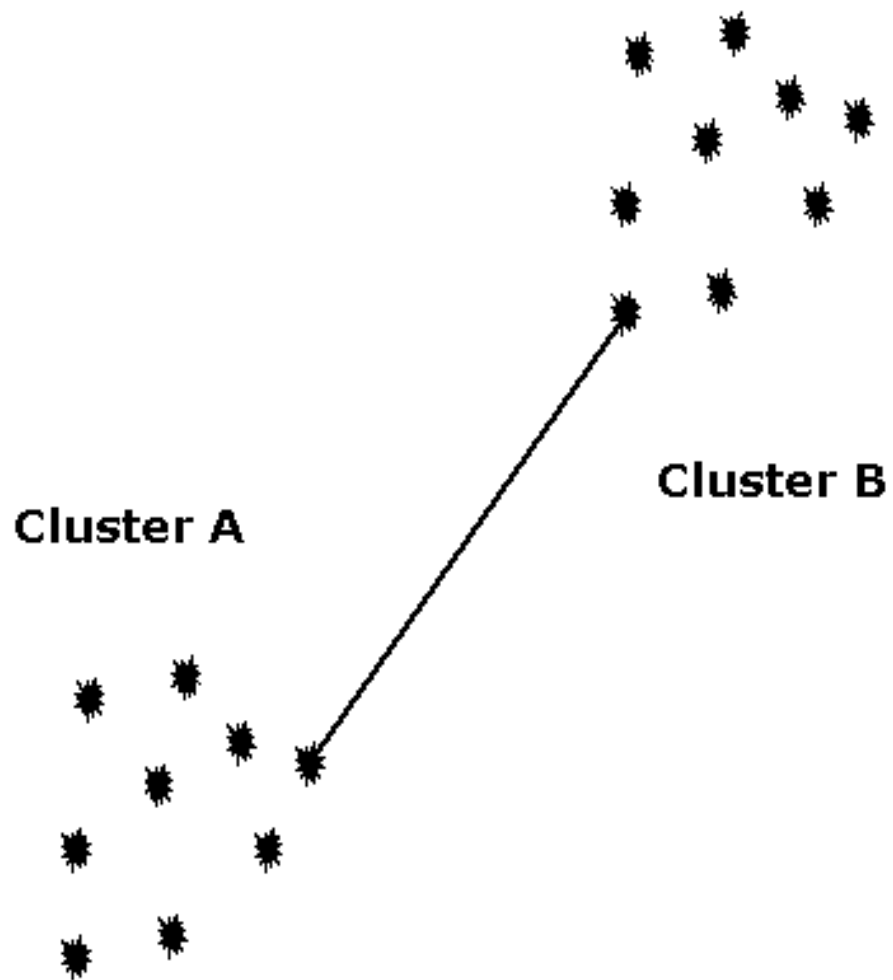


Figure 6: SingleLinkage

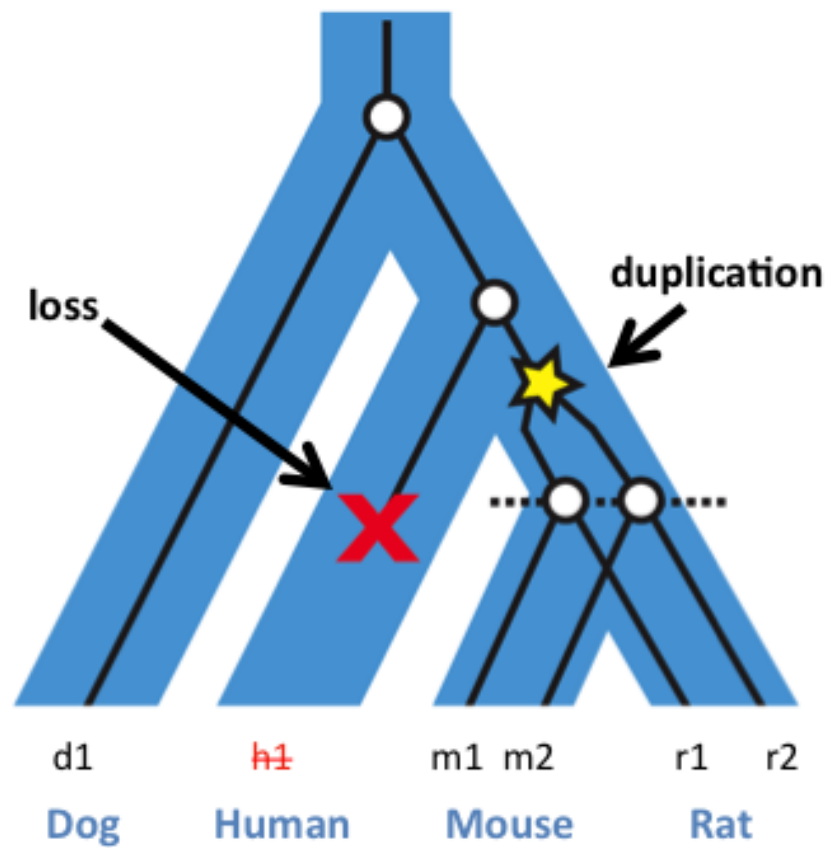


Figure 7: orthologsloss

## Issues

## Tools to go after Orthologous and Paralogous Sequences

- [OrthoFinder](#)

### Steps to build orthologs on cluster

We will take 3 datasets of annotated Cyanobacteria, download and run analysis to generate Ortholog table.

```
#!/usr/bin/bash
#SBATCH --ntasks 16 --mem 8G -p short
module load ncbi-blast
module load orthofinder
module load miniconda2
CPU=8

mkdir -p cyanobacteria
cd cyanobacteria
curl -L -O ftp://ftp.ensemblgenomes.org/pub/bacteria/release-45/fasta/bacteria_10_collection
curl -L -O ftp://ftp.ensemblgenomes.org/pub/bacteria/release-45/fasta/bacteria_0_collection
curl -L -O ftp://ftp.ensemblgenomes.org/pub/bacteria/release-45/fasta/bacteria_4_collection

# uncompress files and name them all *.fasta
for file in *.fa.gz
do
    m=$(basename $file .pep.all.fa.gz)
    pigz -dc $file > $m.fasta
done

cd ..

orthofinder -a $CPU -f cyanobacteria
```

### Ortholog results

The output file by default will be the date of the analysis. Opening the file `cyanobacteria/Results_XXX/Orthogroups.txt` but I made a [folder](#) in the examples you can look over. Here's one [table](#)

### Format

GroupName\tSp1\_Gene1, Sp1\_Gene2\tSp2\_Gene1, Sp2\_Gene2\tSp3\_Gene1, Sp3\_Gene2

Cyanobacterium\_aponinum\_pcc\_10605.ASM31767v1      Nostoc\_punctiforme\_pcc\_73102.ASM2002v1

OG0000000      EKQ66605, EKQ66611, EKQ66662, EKQ66782, EKQ66954, EKQ66984, EKQ67084, EKQ67433, EKQ67590, EKQ67680, EKQ67799, EKQ67807, EKQ67983, EKQ68026, EKQ68032, EKQ68054, EKQ69279, EKQ69300, EKQ69345, EKQ69368, EKQ69506, EKQ69549, EKQ69629, EKQ69630, EKQ69655, EKQ70786, EKQ70840, EKQ70870, EKQ70894, EKQ71088, EKQ71090, EKQ71265, EKQ71335

OG0000001      AFZ52442, AFZ54265, AFZ54640      ACC78968, ACC78978, ACC79054, ACC79090, ACC79138, ACC81797, ACC82091, ACC82628, ACC82978, ACC83035, ACC83215, ACC83711, ACC84528, ACC84844, ACC869971, EKQ69995, EKQ70003, EKQ70556, EKQ70833, EKQ71286

OG0000002      AFZ55137      ACC79344, ACC80485, ACC80595, ACC82143, ACC82836, ACC82962, ACC83844, ACC84972, ACC84974, ACC84981, ACC84982, ACC84983, ACC85032 EKQ66950, EKQ67597, EKQ67615, EKQ69997

OG0000003      AFZ53198      ACC78875, ACC78976, ACC79256, ACC79524, ACC79759, ACC80145, ACC80528, ACC82769, ACC83025, ACC83081, ACC83457, ACC83602, ACC83721, ACC83749, ACC84422, ACC85331

OG0000004      ACC80422, ACC80525, ACC80662, ACC80851, ACC80857, ACC80914, ACC81440, ACC81516, ACC83981, ACC84622, ACC84732, ACC85457      EKQ66830, EKQ66911, EKQ67039, EKQ67311, EKQ69997

OG0000005      AFZ52318, AFZ52611, AFZ52613, AFZ52925, AFZ52973, AFZ53626, AFZ53840, AFZ53841, ACC82559, ACC83603, ACC83674, ACC85005, ACC85009 EKQ67574, EKQ67809, EKQ69976

OG0000006      AFZ52319, AFZ53394, AFZ54017, AFZ54472      ACC79360, ACC79745, ACC79853, ACC80832, ACC81478, EKQ67551, EKQ67724, EKQ67810, EKQ68266

OG0000007      AFZ53704, AFZ54461, AFZ54462      ACC79786, ACC80242, ACC80282, ACC80538, ACC80768, ACC81422, EKQ68369, EKQ70142, EKQ70145, EKQ71300

The tool also generates [summary statistics](#) we can look through.

Could write a script to turn this into a table or use the [summary count table](#) provided.

| ORTHOLOG_GRP | SP1 | SP2 | SP3 |
|--------------|-----|-----|-----|
| ORTHO_0001   | 10  | 5   | 0   |
| ORTHO_0002   | 1   | 1   | 0   |